

CAGAYAN DE ORO LUMBIA, Philippines

WMO: 987470

Lat: 8.409N

Lon: 124.612E

Elev: 617

StdP: 14.37

Time Zone: 8.00 (E08)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	69.7	71.0	64.9	94.3	71.9	66.6	100.1	73.4	14.2	80.7	11.8	81.7	3.8	180	0.265

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.3	91.8	79.4	90.8	79.3	89.8	79.2	81.4	88.7	80.8	88.2	80.4	87.7	5.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	155.3	86.5	78.8	152.7	86.2	78.2	149.9	85.7	45.7	89.0	45.1	88.6	44.6	88.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.2	7.6	64.4	95.7	4.6	2.1	61.1	97.2	58.4	98.4	55.8	99.6	52.4	101.2		
(5)			62.6	83.7	3.2	1.3	60.2	84.6	58.3	85.4	56.5	86.1	54.1	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	78.3	78.7	79.8	81.5	82.2	81.2	80.6	81.0	80.8	80.6	80.1	79.5		
	DBStd	2.04	1.95	1.99	1.89	1.89	1.77	1.65	1.61	1.77	1.59	1.31	1.53	1.65		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11081	877	803	925	945	999	935	949	961	923	948	903	913		
	CDD65	5606	412	383	460	495	534	485	484	496	473	483	453	448		
	CDH74	48267	2897	2897	4021	4990	5350	4290	4067	4329	3965	4102	3802	3557		
	CDH80	16185	700	828	1332	1930	2070	1492	1373	1566	1389	1338	1184	983		
(14) Wind	WSAvg	3.1	3.3	3.5	3.5	3.5	3.3	2.9	2.8	2.9	3.0	2.9	3.0	3.3		
(15) Precipitation	PrecAvg	79.1	4.8	3.9	3.1	3.3	6.5	10.1	10.3	8.7	9.8	7.5	6.1	5.7		
	PrecMax	102.1	17.0	10.3	6.2	8.6	10.3	17.4	14.0	12.1	14.2	13.2	20.3	15.6		
	PrecMin	55.6	0.8	0.6	0.6	0.7	2.1	5.4	7.1	5.2	5.1	4.1	1.5	0.6		
	PrecStd	13.5	3.8	2.7	1.6	2.6	2.2	3.0	2.1	2.1	2.5	2.3	4.5	3.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.8	89.7	91.3	92.7	93.4	91.8	91.7	92.8	92.0	90.7	90.0	89.5		
		MCWB	77.7	77.7	77.4	79.3	79.7	80.2	79.2	79.7	79.9	79.3	79.2	79.2		
	2%	DB	87.0	88.0	89.5	91.2	91.6	89.9	89.6	90.3	89.9	89.2	88.6	87.9		
		MCWB	77.6	77.4	78.0	79.1	79.6	79.8	79.6	79.4	79.5	79.3	78.9	78.8		
	5%	DB	85.7	86.5	88.1	89.8	90.0	88.7	88.1	88.9	88.5	87.9	87.5	86.6		
		MCWB	77.3	77.2	77.9	78.6	79.6	79.5	79.2	79.2	79.1	79.1	78.7	78.5		
	10%	DB	84.1	85.2	86.6	88.5	88.8	87.4	86.8	87.5	87.2	86.6	86.1	85.2		
		MCWB	76.9	77.0	77.6	78.4	79.3	79.2	78.8	78.9	78.7	78.7	78.5	78.1		
	0.4%	WB	79.9	80.0	80.8	81.7	81.9	81.8	81.5	81.7	81.6	81.0	81.0	80.8		
		MCDB	86.1	86.6	87.7	89.1	90.0	89.3	88.5	89.4	89.0	87.6	87.3	86.8		
	2%	WB	78.8	79.0	79.6	80.6	81.0	80.8	80.6	80.7	80.5	80.3	80.1	79.8		
		MCDB	84.8	85.4	86.6	88.6	89.0	88.2	87.6	88.1	88.1	86.9	86.5	86.0		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	78.1	78.2	78.8	79.7	80.4	80.1	79.8	79.9	79.7	79.7	79.4	79.1		
		MCDB	83.7	84.4	85.9	87.7	88.4	87.3	86.7	87.3	87.1	86.3	85.8	85.2		
	10%	WB	77.3	77.4	78.1	78.9	79.7	79.3	79.0	79.1	78.9	79.0	78.8	78.4		
		MCDB	82.7	83.4	84.9	87.0	87.5	86.4	85.8	86.5	86.2	85.6	85.1	84.2		
	5% DB	MDBR	11.4	12.5	13.9	15.0	14.3	13.7	13.5	14.2	14.2	13.7	13.0	12.3		
		MCDBR	14.2	15.4	16.2	16.4	15.7	15.0	15.3	15.9	15.7	15.1	14.8	14.2		
(36) Mean Daily Temperature Range	5% WB	MCWBR	7.5	8.2	8.2	8.0	7.3	7.8	8.2	8.3	8.3	8.1	8.1	7.8		
		MCDBR	12.6	13.4	14.3	15.3	14.7	14.3	14.2	15.1	14.9	14.1	13.8	13.2		
	5% WB	MCWBR	7.3	7.7	8.0	8.4	7.6	8.0	8.3	8.7	8.5	8.2	8.2	7.8		
		MCDBR	12.6	13.4	14.3	15.3	14.7	14.3	14.2	15.1	14.9	14.1	13.8	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.397	0.394	0.396	0.408	0.418	0.408	0.407	0.427	0.426	0.434	0.405	0.398		
	taud		2.512	2.501	2.501	2.465	2.456	2.488	2.488	2.413	2.409	2.386	2.489	2.505		
	Ebn at Noon		291	294	293	285	277	277	278	277	280	279	286	288		
	Edh at Noon		34	35	36	36	36	34	35	38	38	39	35	34		
(44) All-Sky Solar Radiation	RadAvg		1315	1545	1740	1862	1725	1596	1574	1636	1624	1563	1465	1374		
	RadStd		169	170	134	101	103	76	86	108	108	101	124	119		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.40	N/A	N/A	+0.79	+0.59	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page